# Pierre Guilbert

Computer Vision, Machine Learning & Mathematical Modelling
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Engineer with 10 years of experience in Computer Vision, Machine Learning, and Mathematical Modelling, with a strong interest in differential geometry and nonlinear optimization.

Passionate about applying advanced mathematics to real-world challenges, I enjoy combining technical expertise and creative problem-solving through collaboration and intellectual exchange.

## Experience

**Genius Sports** — Senior Computer Vision Engineer 2019 - Present

*Lausanne*

- Camera modelling and calibration: keypoint detection, bundle adjustment, and optical SLAM.
- Human pose estimation: body keypoint detection, triangulation, and trajectory optimization based on physical laws and optical measurements.
- Geometric and physical parameter estimation from multi-view images: 3D reconstruction, Gaussian splatting, ball spin, and ball deformation.
- Ad-tech: auction price optimization and user click-through prediction.

**Kitware** — Computer Vision Engineer 2016 - 2019

*Lyon*

- Software engineering for VeloView: lidar, multi-sensor fusion, and point-cloud processing.
- Lidar-based SLAM: design and implementation of a lidar-based SLAM system inspired by the LOAM algorithm.
- Contributions to open-source projects: VTK, ParaView, and Ceres Solver.

**CNRS** — Image Processing Research Assistant 2015 - 2016

*Saint-Étienne*

- Research in image and signal processing using the logarithmic image processing (LIP) model.
- Neural network training framework using LIP operations.

## Skills

**Mathematical Modelling**: Physics, PDEs, variational calculus, nonlinear optimization

**Computer Vision**: Multi-view geometry, bundle adjustment, SLAM, pose estimation

**Machine Learning**: Design and training of discriminative and generative models (classification, segmentation, regression, density estimation) with PyTorch

**Mathematics**: Differential geometry, topology, measure theory, algebra

**Programming**: C++ (Eigen, Ceres, OpenCV, PCL), Python, Git

## Education

**Graduate studies in Mathematics & Engineering** — Mines Saint-Étienne 2016

**MSc Computer Graphics & Signal Processing** — Université Lyon 1 2015

**MEng Electrical Engineering** — Université Lyon 1 2015

**BSc Mathematics, Physics & Chemistry** — CPGE Champollion 2012

## Personal Interests

**Philosophy & Literature**: Exploring morality, ethics, consciousness, and the nature of reality

**Stock market investment**: Long-term stock picking with no intention to outperform the market, but to learn from it

**Powerlifting**: Strength training in bench press, deadlift, and squat

**Mathematics**: Exploring the state of the art, writing research notes and short mathematical treatises